

The Distributive Property

Lesson 6-5

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$3(x + 2)$$

$$3x + 6$$

$3(4 + 5) = 3 \times 4 + 3 \times 5 = 12 + 15 = 27$ — the 3 gets 'distributed' to both the 4 and the 5

Distributive Property

Write the definition:

$$3 \times 4 = 12$$

3 and 4 are factors

In $3(x + 2)$, the 3 is the factor outside the parentheses that multiplies each term inside

Factor

Write the definition:

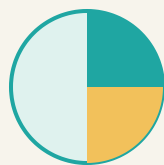
$$3(x + 2)$$

$$3x + 6$$

Expand $2(n + 6)$: multiply $2 \times n = 2n$ and $2 \times 6 = 12$, so $2(n + 6) = 2n + 12$

Expand

Write the definition:



$\frac{3}{4}$

$3(x + 4)$ and $3x + 12$ always give the same answer: when $x = 2$, both = 18; when $x = 10$, both = 42

Equivalent

Write the definition:


$$4x$$

coefficient = 4

In $6x + 15$, the coefficient of x is 6 — it came from distributing in $3(2x + 5)$

Coefficient

Write the definition:


$$3x + 2x = 5x$$

same variable → combine

$4x + 2x = 6x$ (combine coefficients); $4x + 2y$ cannot be combined (different variables)

Like terms

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can use the distributive property to expand and factor expressions.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Distributive Property

Factor

Expand

Equivalent

Coefficient

Like terms

- 1 The rule that $a(b + c) = ab + ac$ is the _____.
- 2 To rewrite a sum as a product, like $3x + 12 = 3(x + 4)$, is to _____.
- 3 To multiply out a product over a sum, like $3(x + 4) = 3x + 12$, is to _____.
- 4 Two expressions that always have the same value are _____.
- 5 A number multiplied by a variable is a _____.
- 6 Terms with the same variable part are _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: Expand $6(n + 3)$ using the distributive property.

A. $6n + 18$

B. $6n + 3$

C. $n + 18$

D. $6n + 9$

Step 1 $6(n + 3) = 6 \times n + 6 \times 3 = 6n + 18.$

 **Answer:** A. $6n + 18$

 We do — together

Solve this one with your class using the same steps.

Problem: Expand $4(5 - 2)$ using the distributive property.

A. 12

B. $20 - 2$

C. 9

D. 22

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Expand $3(x + 4)$ using the distributive property.

A. $3x + 12$

B. $3x + 4$

C. $x + 12$

D. $7x$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Light Cue 2: Expand $5(2 + n)$ to set two spotlight banks.

A. $10 + 5n$

B. $10 + n$

C. $7 + 5n$

D. $10 + 5$

Show your work:

2. Light Cue 3: Expand $4(a + 2)$ for the back-row beam.

A. $4a + 8$

B. $4a + 2$

C. $a + 8$

D. $4a + 6$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Use the distributive property to explain why 6×98 can be calculated as $6 \times 100 - 6 \times 2 = 588$. Then create your own example of using the distributive property to make multiplication easier.

Sentence starter: $6 \times 98 = 6(100 - 2) = 6 \times \underline{\quad} - 6 \times \underline{\quad} = \underline{\quad} - \underline{\quad} = \underline{\quad}$. *My example:* $\underline{\quad} \times \underline{\quad} = \underline{\quad}(\underline{\quad} \pm \underline{\quad}) = \underline{\quad}$.

Show your work:

Reflect — Exit Ticket

Which expression is equivalent to $7(x + 3)$?

- A. $7x + 21$
- B. $7x + 3$
- C. $x + 21$
- D. $7x + 10$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Distributive Property
2. **Notes 2:** Factor
3. **Notes 3:** Expand
4. **Notes 4:** Equivalent
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **We Try (notes):** A. $12 - 4(5 - 2) = 4 \times 5 - 4 \times 2 = 20 - 8 = 12.$
8. **You Try (notes):** A. $3x + 12 - 3(x + 4) = 3 \times x + 3 \times 4 = 3x + 12.$
9. **Try It 1:** A. $10 + 5n$ — *Distribute the 5 to each term: $5 \times 2 + 5 \times n = 10 + 5n.$*
10. **Try It 2:** A. $4a + 8$ — *Distribute the 4 to each term: $4 \times a + 4 \times 2 = 4a + 8.$*
11. **Exit Ticket:** A. $7x + 21 - 7(x + 3) = 7 \times x + 7 \times 3 = 7x + 21.$